

CO4_AT1 _QUESTIONS

INDUSTRIAL PROBLEM SOLVING QUESTIONS

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30) A factory-floor control panel needs a perfectly uniform grid of 12 identically-sized indicator buttons, 3 rows by 4 columns. The buttons should resize together proportionally when the operator maximizes the window. Which layout manager guarantees uniformity and why would FlowLayout fail to preserve the grid structure on resize?

A) AIM:

To create a Java GUI containing 12 identically-sized indicator buttons arranged in a 3×4 uniform grid and demonstrate how GridLayout maintains the grid structure when the window is resized.

PROCEDURE:

- 1) Create a Java Swing application using JFrame.
- 2) Create a JPanel and set its layout manager to GridLayout.
- 3) Specify **3 rows and 4 columns** in the GridLayout.
- 4) Create 12 buttons representing the indicator buttons.
- 5) Add all 12 buttons to the panel.
- 6) Add the panel to the frame.
- 7) Set the frame size and make it visible.
- 8) Resize or maximize the window.
- 9) Observe that all buttons remain in a uniform **3×4 grid** and resize proportionally. Compare this with FlowLayout, which arranges components sequentially and may wrap them differently when the window size changes.

CODE:

```
import javax.swing.*;
import java.awt.*;

public class FactoryControlPanel {

    public static void main(String[] args) {

        JFrame frame = new JFrame("Factory Floor Control Panel");

        // Create a 3 x 4 uniform grid
        JPanel panel = new JPanel(new GridLayout(3, 4, 5, 5));

        // Create 12 indicator buttons
        for (int i = 1; i <= 12; i++) {
            JButton button = new JButton("Indicator " + i);
            panel.add(button);
        }

        frame.add(panel);

        frame.setSize(600, 400);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLocationRelativeTo(null);
        frame.setVisible(true);
    }
}
```

INPUT:

Number of Rows : 3
Number of Columns : 4
Number of Buttons : 12

Buttons:

Indicator 1
Indicator 2
Indicator 3
Indicator 4
Indicator 5
Indicator 6
Indicator 7
Indicator 8
Indicator 9
Indicator 10
Indicator 11
Indicator 12

OUTPUT:

Indicator 1	Indicator 2	Indicator 3	Indicator 4
Indicator 5	Indicator 6	Indicator 7	Indicator 8
Indicator 9	Indicator 10	Indicator 11	Indicator 12

RESULT:

The factory-floor control panel was successfully created using **GridLayout**. The 12 indicator buttons are arranged uniformly in **3 rows and 4 columns**. When the window is resized or maximized, all buttons resize proportionally while maintaining the grid structure.